The Sounding Page

Profiles, indices, time–height sections and radiosonde checks from aircraft data

Otso Laakso (OH2GAX)
Tuusula, Finland · near Helsinki-Vantaa airport (EFHK)

In short. The Sounding page turns the temperatures and winds that MODE-S Wind derives from aircraft transponder replies into the tools a meteorologist uses for upper-air data. Five buttons choose the view. Area · now, Area · hour and Flight draw one vertical profile on a Skew-T diagram, with an Indices panel of stability indices and characteristic levels and a table of the layer values; the Jokioinen radiosonde can be drawn beside it, and its indices are computed by the same code for a fair comparison. Time–height puts the hourly profiles side by side, so that fronts, inversions, jet streams, sinking air and the daily cycle are seen as they pass over the receiver. Sonde check measures how well the aircraft profile agrees with the radiosonde, and how quickly a sounding goes out of date.
This guide explains each view and the physics behind what it shows. How to read the Skew-T diagram itself — the lines, the automatic Analysis, the airport low-level profile — is explained in the companion guide *Reading the Atmosphere Profile (Skew-T)*; here it is only summarised. The examples come from 16–19 October 2026: a cold front on 17 October and a strong high-pressure area (QNH 1040 hPa) behind it.

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1 The page at a glance

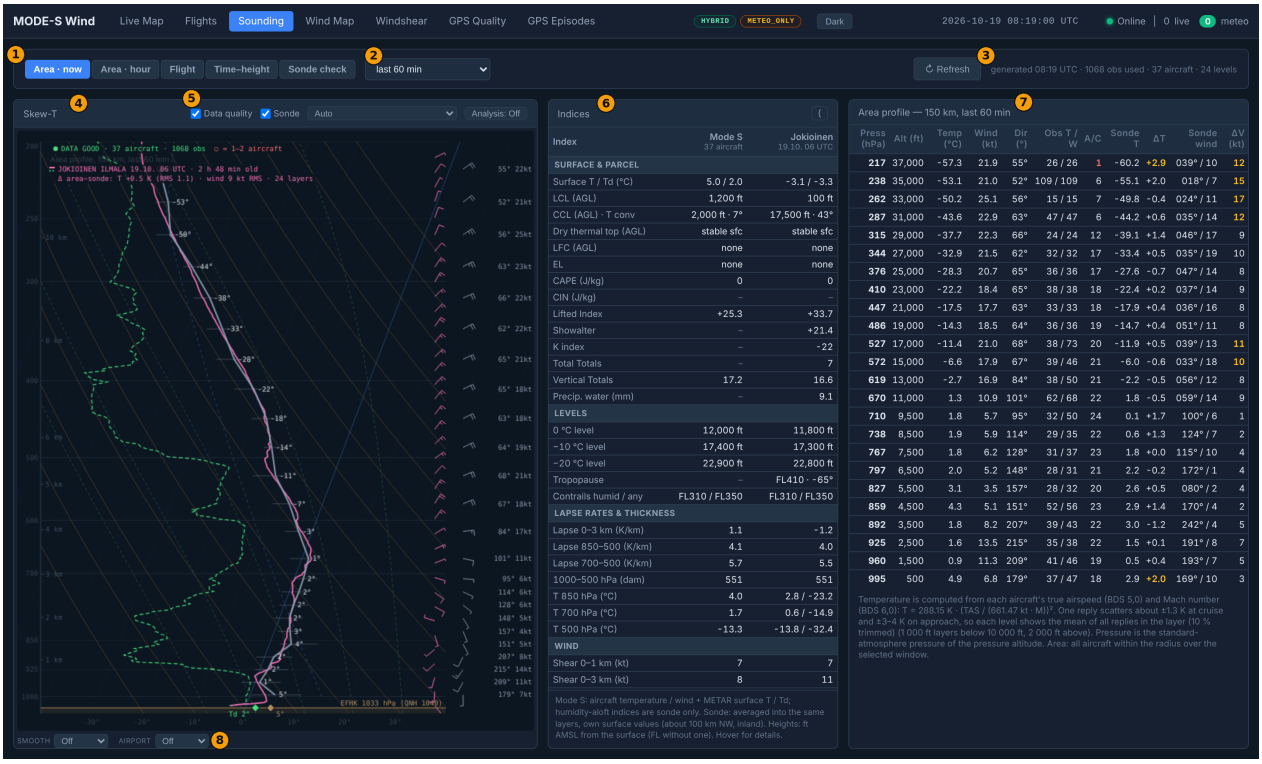


Figure 1. The Sounding page, Area · now, 19 October 2026 07:19–08:19 UTC, with Sonde on. (1) view buttons, (2) the controls of the view (here the time window), (3) refresh and summary: observations, aircraft and layers used, (4) Skew-T diagram: grey Mode S area profile, pink radiosonde, green dashed sonde dewpoint, (5) Data quality, Sonde with the sounding selector, Analysis, (6) Indices panel: Mode S and Jokioinen side by side, (7) level table with the sonde columns, (8) Smooth and Airport. Surface values from the METAR stored for 07:50 UTC (5/2 °C, QNH 1040).

The five buttons at the top left ((**1**) in Fig. 1) choose what the page shows; the choice is remembered in the browser.

Area · now — the live area profile: all aircraft within 150 km of the receiver over the last 30 or 60 minutes, 2 or 3 hours (default 60 minutes), with the current METAR at the ground.

Area · hour — the *hourly profile archive*: one area profile stored for every completed UTC hour, kept indefinitely, also after the observations themselves have been purged. Pick a date and an hour; the list shows how many observations each hour has. The METAR of that hour (temperature, dewpoint, QNH, lowest cloud) is stored with it (from 11 October 2026).

Flight — the profile of one climbing or descending aircraft, chosen from a list of recent flights with a large enough altitude range (or opened with the Sounding button of the Flights page).

Time–height — the hourly profiles side by side over 24 h to 14 days (Section 5).

Sonde check — the automatic verification against the radiosonde (Section 6).

The first three views share the same layout: the Skew-T on the left, the Indices panel in the middle and the level table on the right. On narrow screens the table moves below the diagram, and the ⟨ button hides the Indices panel (a slim Indices tab brings it back).

2 Where the profile comes from

Aircraft as weather sensors. Airliners answer the interrogations of ground radars with *Enhanced Surveillance* reports. Two of them carry the airspeeds and directions of the aircraft: **BDS 5,0** (true track, groundspeed, true airspeed TAS, roll angle) and **BDS 6,0** (magnetic heading, Mach number, indicated airspeed). From each pair MODE-S Wind computes

- the **air temperature**, because the ratio of true airspeed to Mach number is the speed of sound, which depends only on temperature: $T = 288.15\,\left(\mathrm{TAS}/(661.47\,\mathrm{kt}\cdot M)\right)^2\,\mathrm{K}$;
- the **wind**, as the difference between the ground vector (track, groundspeed) and the air vector (true heading, TAS) — the wind triangle. The heading is corrected for magnetic declination and for each airframe's own heading offset, which the system learns.

A single reply pair scatters by about ±1.3 K at cruise and 3–4 K at approach speeds, because Mach is sent in steps of 0.004 and TAS in steps of 2 kt. Averaging many replies removes most of this; no per-airframe temperature calibration is needed (the README explains why).

Pressure altitude is pressure. The altitude an aircraft reports is its *pressure altitude*: the height at which the measured static pressure would occur in the International Standard Atmosphere (ISA), referenced to 1013.25 hPa. FL300 is therefore always the 301 hPa surface, whatever the weather. The page uses this directly as the vertical coordinate: each layer's pressure is the ISA pressure of its pressure altitude. True heights above the sea differ from pressure altitude with the QNH and with the temperature of the air column; the Indices panel computes true heights from the METAR surface pressure (Section 4).

The area profile. Observations within 150 km of the receiver are grouped into layers 1 000 ft deep up to 10 000 ft and 2 000 ft deep above (to 46 000 ft). For each layer the **temperature** is the 10 % trimmed mean (the warmest and coldest tenth removed) and the **wind** the vector mean. A daytime hour typically has 25–70 different aircraft, about 20 per layer; around midnight UTC only 2–8 aircraft fly in the area. The profile is thus an *area and time average*: 300 km across and 60 minutes long. A radiosonde is the opposite: one point, measured within about an hour, but at full vertical resolution and with humidity.

What is not measured. Aircraft report no humidity. The only moisture value is the METAR dewpoint at the airport, so the Mode S column of the Indices panel cannot give indices that need humidity aloft.

3 One profile: Area, hour or flight

3.1 The Skew-T diagram in brief

The diagram is the same as the Atmosphere Profile of the Live Map, only larger. Pressure is the vertical axis (logarithmic, 1000 hPa at the bottom), temperature runs along lines tilted 45° to the right. The grey curve is the Mode S temperature with the layer values, the short horizontal bars the 10–90 % range of the replies; wind barbs and their direction and speed are on the right. Faint brown lines are dry adiabats, dashed green lines moist adiabats, the bold blue diagonal is 0 °C and the dashed blue line the ISA. At the bottom the brown ground line is the station pressure from the METAR QNH, with the METAR temperature and dewpoint.

The controls work as on the Live Map and share their settings with it:

- Data quality: a badge with the number of aircraft and observations (`DATA GOOD` ≥ 20 aircraft per hour of window, `FAIR` 8–19, `LOW` < 8), and red rings on layers measured by only one or two aircraft;
- Analysis: Off / Levels / Stability / Turbulence / Advection / Full — tropopause, isotherms, jets, inversions, stability strip, turbulence risk, warm and cold advection, contrail levels, thermal top and surface parcel;
- Smooth: a running mean of 3 or 5 layers on the temperature curve;
- Airport: the low-level profile from the landings and departures at EFHK (200 ft bands) of the shown period — the selected window, that hour, or the flight's time ±15 minutes.

3.2 The radiosonde beside the aircraft

The Finnish Meteorological Institute launches a radiosonde at Jokioinen (about 100 km north-west of the receiver, inland) at about 23:30, 05:30 and 11:30 UTC for the 00, 06 and 12 UTC soundings. MODE-S Wind fetches them from FMI open data as they appear — often while the balloon is still climbing — and stores them. With Sonde ticked the sounding is drawn in pink, its dewpoint as a dashed green curve and its winds as pink barbs, and a label gives its age and the mean difference area − sonde.

The Auto setting of the sounding selector chooses: in Area · now the newest sounding; in Area · hour and Flight the newest one launched by the end of that hour or flight (at most 36 h older), with its age counted to the middle of the hour or flight. Any stored sounding can also be picked by hand, for example the one launched *after* an hour, to see how the atmosphere had changed. Partial soundings show their top (`· to 350 hPa`); FMI's open-data copy of a flight often ends at 250–550 hPa.

The two profiles never agree exactly, and the differences are informative. In calm weather (Fig. 1) they agree within about 1 K and 10 kt through most of the column. When a front lies between Jokioinen and Helsinki they can differ by several degrees (Fig. 2).

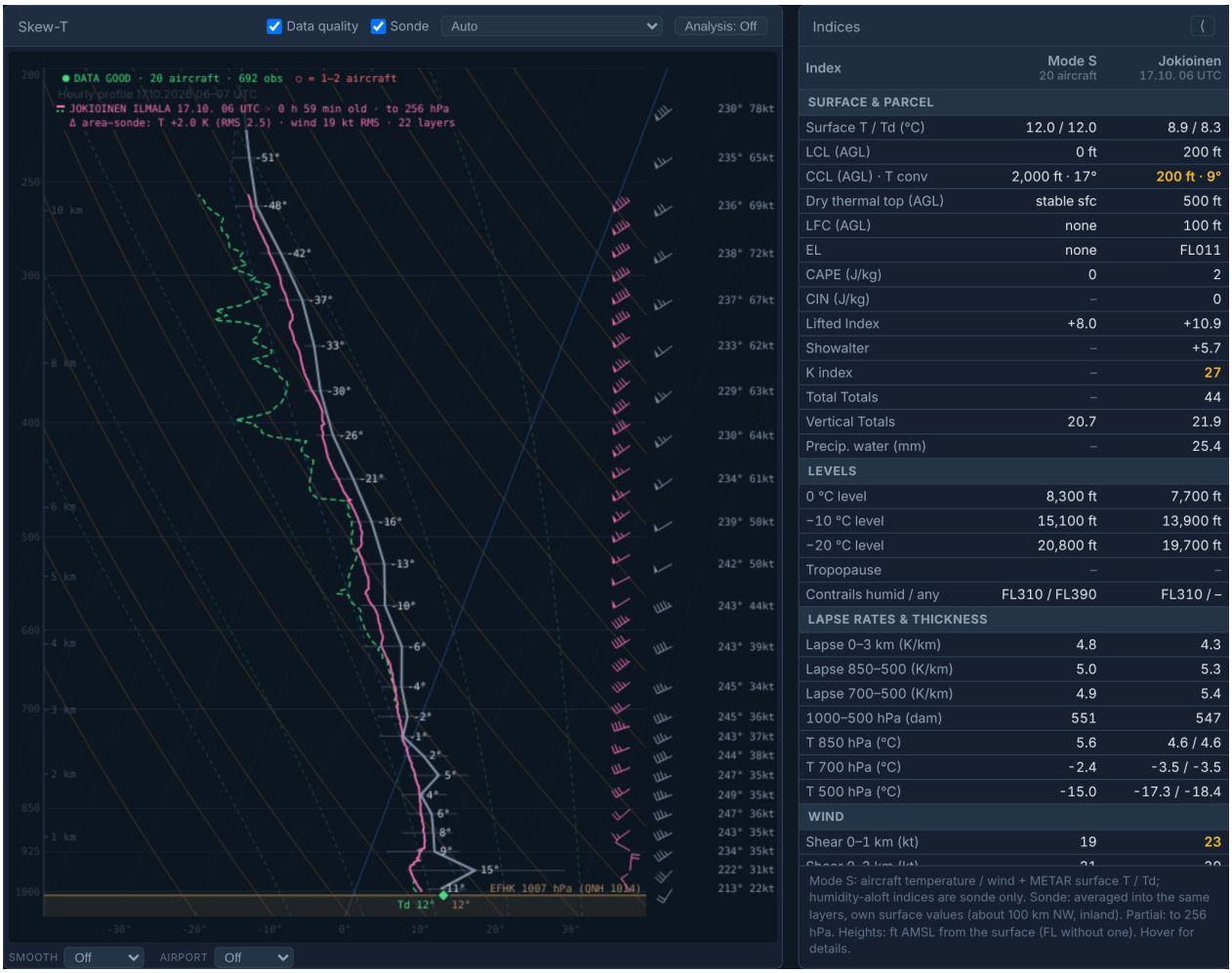


Figure 2. Area · hour 17 October 2026 06–07 UTC with the 06 UTC Jokioinen sounding. Over Helsinki the aircraft still measure the warm, south-westerly air ahead of a cold front: 15 °C and 222°/31 kt at 1 500 ft above a cooler, saturated surface layer (METAR 12/12, BKN006). At Jokioinen, already behind the front, the lowest 2 000 ft are 3–8 K colder with a northerly wind (357–006°), while above about 3 000 ft both show the same south-westerly flow. The whole column is warmer over Helsinki: 1000–500 hPa thickness 551 against 547 dam, about 2 K.

This is a textbook picture of a cold front: the cold air advances as a shallow wedge under the warm air, so at a given moment the front is already past at the ground at Jokioinen but not yet over Helsinki, and aloft both sites are still in the same air. The front reached EFHK at about 08–09 UTC (the wind at 1 000 ft turned from 244° to 353°, the METAR temperature fell from 12 to 9 °C). Section 5 shows how the same front looks in the time–height section.

3.3 A single flight

Flight draws the replies of one aircraft as it climbs out or descends, with its own wind and temperature layers (Fig. 3). It is a single instrument at a single time — the closest thing to a radiosonde the aircraft data offer — but every layer rests on one to six replies of one airframe, so the zigzags of 2–3 K are mostly Mach and TAS steps rather than real structure. Compare it with the area profile of the same hour before reading anything into a single bend. A flight's profile also has gaps where the radar happened not to interrogate the aircraft; the note under the Indices panel lists them, and values above a gap are less certain. A flight has no METAR, so the surface-based indices are left out.

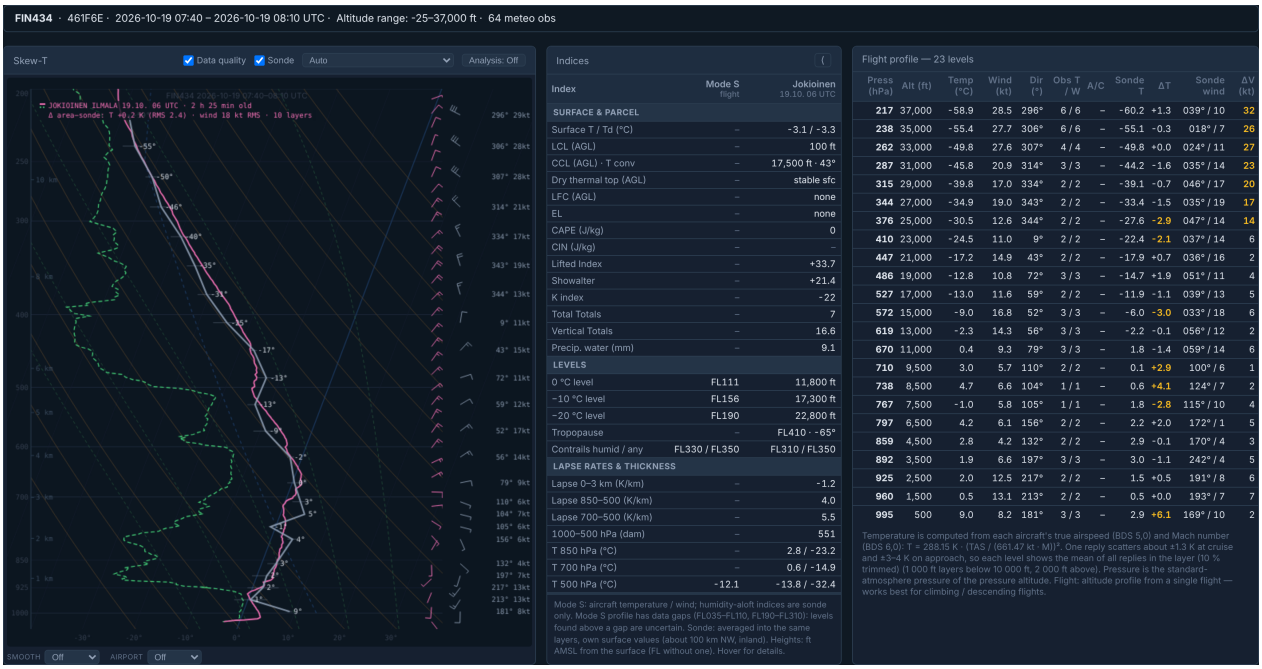
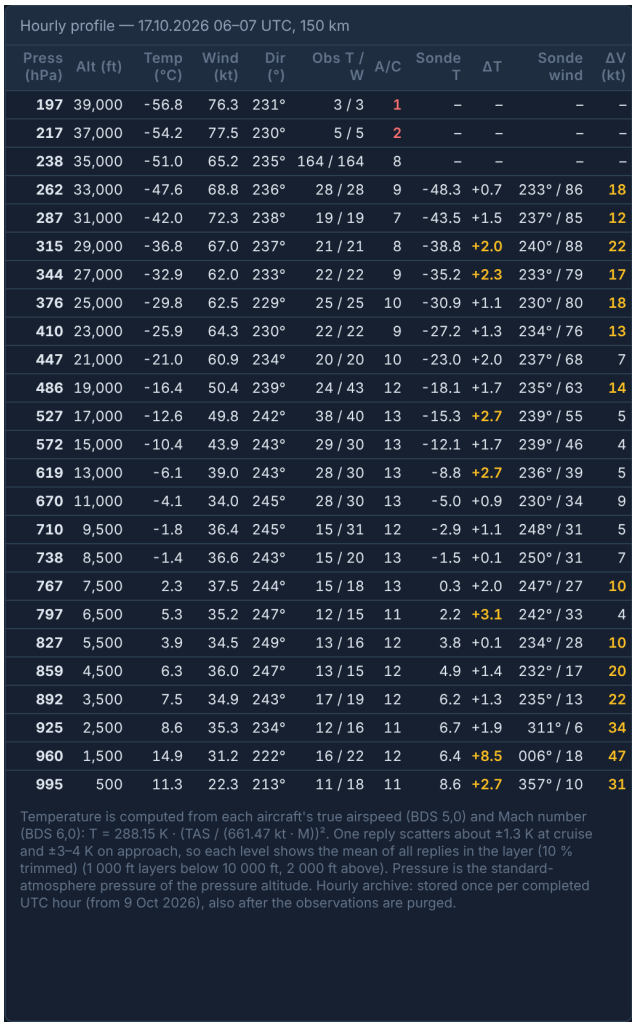


Figure 3. Flight: a departure on 19 October 2026 07:40–08:10 UTC with the 06 UTC sounding. Both show the nearly isothermal layer of 0 to +5 °C between about 2 000 and 11 000 ft; the aircraft's single replies zigzag by 2–4 K around it. Above FL250 the aircraft measured north-westerly 20–28 kt where the sonde, 100 km away and two hours earlier, had north-easterly 7–19 kt: under the high the winds were light and variable, so small differences in place and time matter.

3.4 The level table

The table lists every layer from the top down, as in the diagram: ISA pressure, layer centre (pressure altitude), temperature (hover: 10–90 % range), wind speed (hover: vector standard deviation) and direction, the number of replies for temperature and wind, and the number of different aircraft (A/C, red when only one or two). With Sonde on, four columns follow: the sonde temperature at the layer's pressure, ΔT (layer − sonde, amber beyond 2 K), the sonde wind and ΔV, the length of the vector difference (amber beyond 10 kt). In Fig. 4 the front is the column of large ΔT and ΔV in the lowest 3 000 ft.



Hourly profile — 17.10.2026 06–07 UTC, 150 km

Press (hPa)	Alt (ft)	Temp (°C)	Wind (kt)	Dir (°)	Obs T / W	A/C	Sonde T	ΔT	Sonde wind	ΔV (kt)
197	39,000	-56.8	76.3	231°	3 / 3	1	–	–	–	–
217	37,000	-54.2	77.5	230°	5 / 5	2	–	–	–	–
238	35,000	-51.0	65.2	235°	164 / 164	8	–	–	–	–
262	33,000	-47.6	68.8	236°	28 / 28	9	-48.3	+0.7	233° / 86	18
287	31,000	-42.0	72.3	238°	19 / 19	7	-43.5	+1.5	237° / 85	12
315	29,000	-36.8	67.0	237°	21 / 21	8	-38.8	+2.0	240° / 88	22
344	27,000	-32.9	62.0	233°	22 / 22	9	-35.2	+2.3	233° / 79	17
376	25,000	-29.8	62.5	229°	25 / 25	10	-30.9	+1.1	230° / 80	18
410	23,000	-25.9	64.3	230°	22 / 22	9	-27.2	+1.3	234° / 76	13
447	21,000	-21.0	60.9	234°	20 / 20	10	-23.0	+2.0	237° / 68	7
486	19,000	-16.4	50.4	239°	24 / 43	12	-18.1	+1.7	235° / 63	14
527	17,000	-12.6	49.8	242°	38 / 40	13	-15.3	+2.7	239° / 55	5
572	15,000	-10.4	43.9	243°	29 / 30	13	-12.1	+1.7	239° / 46	4
619	13,000	-6.1	39.0	243°	28 / 30	13	-8.8	+2.7	236° / 39	5
670	11,000	-4.1	34.0	245°	28 / 30	13	-5.0	+0.9	230° / 34	9
710	9,500	-1.8	36.4	245°	15 / 31	12	-2.9	+1.1	248° / 31	5
738	8,500	-1.4	36.6	243°	15 / 20	13	-1.5	+0.1	250° / 31	7
767	7,500	2.3	37.5	244°	15 / 18	13	0.3	+2.0	247° / 27	10
797	6,500	5.3	35.2	247°	12 / 15	11	2.2	+3.1	242° / 33	4
827	5,500	3.9	34.5	249°	13 / 16	12	3.8	+0.1	234° / 28	10
859	4,500	6.3	36.0	247°	13 / 15	12	4.9	+1.4	232° / 17	20
892	3,500	7.5	34.9	243°	17 / 19	12	6.2	+1.3	235° / 13	22
925	2,500	8.6	35.3	234°	12 / 16	11	6.7	+1.9	311° / 6	34
960	1,500	14.9	31.2	222°	16 / 22	12	6.4	+8.5	006° / 18	47
995	500	11.3	22.3	213°	11 / 18	11	8.6	+2.7	357° / 10	31

Temperature is computed from each aircraft's true airspeed (BDS 5,0) and Mach number (BDS 6,0): $T = 288.15\ \mathrm{K} \cdot (\mathrm{TAS} / (661.47\ \mathrm{kt} \cdot M))^2$. One reply scatters about ±1.3 K at cruise and ±3–4 K on approach, so each level shows the mean of all replies in the layer (10 % trimmed) (1 000 ft layers below 10 000 ft, 2 000 ft above). Pressure is the standard-atmosphere pressure of the pressure altitude. Hourly archive: stored once per completed UTC hour (from 9 Oct 2026), also after the observations are purged.

Figure 4. Level table of Fig. 2. The lowest layers differ from the sonde by up to +8.5 K and 47 kt (opposite wind directions across the front); higher up the differences are mostly within ±2 K and 5–20 kt. The top three layers are above the end of the sounding.

4 The Indices panel

The panel lists stability indices and characteristic levels for the shown profile (Mode S) and, with Sonde on, for the radiosonde (Jokioinen). Both columns are computed by the same code, and the sounding is first averaged into the same 1 000 / 2 000 ft layers as the aircraft profile, so the comparison shows measurement differences, not differences in resolution. The sonde column uses the sonde's own surface values; the Mode S column uses the METAR. Hover a label for its definition and a value for details (its pressure, all inversions or turbulence layers, the further crossings of an isotherm, or why a value cannot be given). Amber and red mark values that indicate instability or strong gradients.

Heights are computed *hypsometrically* from the surface point: each layer's thickness follows from its mean temperature (Section A). Parcel levels are given in feet above the ground (AGL), isotherms in feet above mean sea level; without a surface point (a flight) only flight levels are given. Temperatures are not interpolated across a data gap of more than one missing 2 000 ft layer, and the note lists such gaps.

4.1 Surface and parcel

These rows lift the surface air — the METAR temperature and dewpoint — and compare it with the profile. They describe convection that starts from the ground.

LCL Lifting condensation level: rising air cools at the dry adiabatic rate (9.8 K/km) until it reaches its dewpoint and condenses. With a well-mixed boundary layer this is the base of cumulus clouds, roughly 400 ft for each degree of spread $T - T_d$. In Fig. 2 the METAR 12/12 gives an LCL at the ground: the air was saturated (low stratus).

CCL and convective temperature The convective condensation level is where the line of constant moisture through the surface dewpoint meets the profile; the convective temperature is the surface temperature that would carry dry thermals up to it. When the surface reaches it (amber), cumulus from surface heating alone can form.

Dry thermal top The surface temperature lifted along the dry adiabat until it meets the profile: the depth of dry thermals

and of daytime mixing. `stable sfc` means the air above the ground is warmer than surface air lifted to it — a surface inversion.

LFC, EL, CAPE, CIN Above the LCL the parcel follows a moist adiabat. If it becomes warmer than its surroundings (level of free convection, LFC) it rises freely until it is again as cold as the environment (equilibrium level, EL, roughly the top of convective clouds). CAPE is the energy the parcel gains in between (amber ≥ 100, red $\geq 1\,000$ J/kg); CIN is the energy needed to lift it to the LFC.

Lifted Index, Showalter Environment minus parcel temperature at 500 hPa for a surface parcel (LI) or an 850 hPa parcel (Showalter, sonde only). Negative values mean unstable air (amber ≤ -1, red ≤ -5 for LI).

K index, Total Totals, Vertical Totals Classic air-mass thunderstorm indices from the temperatures and dewpoints at 850, 700 and 500 hPa. Only Vertical Totals, $T_{850} - T_{500}$, needs no humidity and is available from the aircraft; values over about 30 mean steep mid-level lapse rates.

Precipitable water All the water vapour of the column, as millimetres of liquid water (sonde only).

Under a high-pressure area, as in Fig. 1, all of these say the same thing: very stable air (Lifted Index $+25$ to $+34$, no LFC, `stable sfc`). The LCL of the Mode S column (1 200 ft) and of the sonde (100 ft) differ because the surface values differ: 5/2 °C at the airport against $-3/-3$ °C at the sonde site, inland after a clear night.

4.2 Levels

0, −10, −20 °C levels The lowest crossing of each isotherm; `(+n)` and the hover list further crossings, which mean a warm or cold layer aloft. The freezing level matters for the type of precipitation and for airframe icing, which is most common in supercooled cloud between 0 and about −15 °C.

Tropopause The WMO definition: the lowest level above 5 km where the lapse rate falls to 2 K/km or less and stays so for at least 2 km. The aircraft data often end at about FL410, so a high tropopause is found only by the sonde (FL410 in Fig. 1).

Contrails The lowest level where a jet engine's exhaust forms a contrail if the air is saturated / at any humidity (Schmidt–Appleman criterion). Contrails persist only in air that is supersaturated with respect to ice, which the aircraft cannot measure.

When the 0 °C level is ill-defined. On 18–19 October the air between the ground and about 11 000 ft was nearly isothermal at 0 to +4 °C, the result of sinking air under the high (Section 5.5). In such a layer the first crossing of 0 °C is decided by a few tenths of a degree, and noise of ±1 K can move it by several thousand feet from one hour to the next (the jumping 0 °C line on 19 October in Figs. 6–9). This is a property of the atmosphere, not an error: the hover list of further crossings shows it.

4.3 Lapse rates, thickness and wind

Lapse rates Temperature fall per km between the surface and 3 km, 850–500 and 700–500 hPa. The ISA has 6.5 K/km; values from 7 (amber) towards the dry adiabatic 9.8 K/km (red from 9) mean steep, potentially unstable layers. Small or negative values mean stable air or inversions.

1000–500 hPa thickness The depth of the layer between the two pressure surfaces, in decametres. By the hypsometric equation it depends only on the layer's mean temperature: 1 dam corresponds to about 0.5 K. 551 dam means a layer mean of about −1.5 °C; about 528 dam is a common rule-of-thumb divide between rain and snow. Thickness is an excellent comparison between the aircraft and the sonde, because it averages the whole lower half of the troposphere: on 14–19 October the two agreed to +0.6 dam on average.

T 850 / 700 / 500 hPa Temperatures at the standard pressure levels used on weather maps (sonde: and dewpoints).

Bulk shear 0–1, 0–3, 0–6 km The vector difference between the wind at that height above the ground and the lowest layer. Strong 0–6 km shear (≥ 40 kt, amber) organises convection when there is any.

Max wind, low-level jet The strongest wind (a jet stream from 60 kt) and a wind maximum below 5 000 ft that is at least 25 kt and 10 kt stronger than above it.

Lowest inversion, turbulence layers From the automatic analysis: the lowest significant inversion with its temperature rise, and layers with a turbulence risk from strong shear and a small Richardson number.

Advection The strongest warm or cold advection, estimated from the turning of the wind with height (the thermal wind, Section 5.4).

The area profile is a 150 km, 60-minute average, so its shear and maximum wind are usually smaller than in the sonde: on 14–19 October the Mode S 0–6 km shear was on average 8 kt lower.

5 The time–height section

A single sounding is a snapshot. The weather is a film: air masses, fronts, troughs and ridges move over the receiver, and the profile changes from hour to hour. The Time–height view shows this film on one picture. It is built from the hourly profile archive: **one column per UTC hour, one row per altitude layer, time running to the right.** The current hour is added live from the observations (dashed frame).

5.1 How to read the picture

- **Height scale.** The vertical axis is the square root of the pressure altitude, which stretches the lowest levels: the 1 000 ft layers below 5 000 ft keep a readable height, while FL300–FL420 are compressed. The left axis gives feet and flight levels, the right axis the ISA pressure.
- **Colour field** (selector): temperature anomaly, temperature change over 3 h, wind speed or lapse rate (below).
- **Barbs**: wind barbs on every second column and layer as space allows, with a dark halo so they read on any colour.
- **Isotherms**: the lowest 0 °C crossing of each hour (solid) and the −10 and −20 °C crossings (dashed, dotted), joined from hour to hour.
- **Markers**: pink triangles at the top mark the Jokioinen launches; day separators carry the weekday.
- **Aircraft strip**: below the plot, the number of aircraft in each hour; red bars (< 8 aircraft, LOW) and amber bars (< 20) warn that the column rests on little data — typically 22–02 UTC. Cells with fewer than 3 replies are drawn faint; empty cells (dark) had no data at all.
- **Hover and click**: the tooltip gives the hour, the layer and its pressure, temperature with anomaly and 3-hour change, lapse rate, wind and aircraft counts (Fig. 5). A click opens that hour on the Skew-T (Area · hour), with its indices and the sonde.
- **Length and end**: 24 h, 48 h, 72 h, 7 or 14 days, ending now or at the end of a chosen UTC day.

Time is (almost) space. Weather systems move, mostly with the wind of the middle troposphere. What passes over the receiver during a day was upstream the day before, so a time–height section is close to a vertical cross-section through the weather system along its direction of motion — with one twist: because time runs to the right, the part of the system that arrives *first* (its leading, downstream side) is on the *left*. A system moving at 20 kt covers about 900 km in 24 hours; the 300 km wide averaging area corresponds to several hours (about 8 hours at 20 kt), so sharp features are smoothed over several columns.

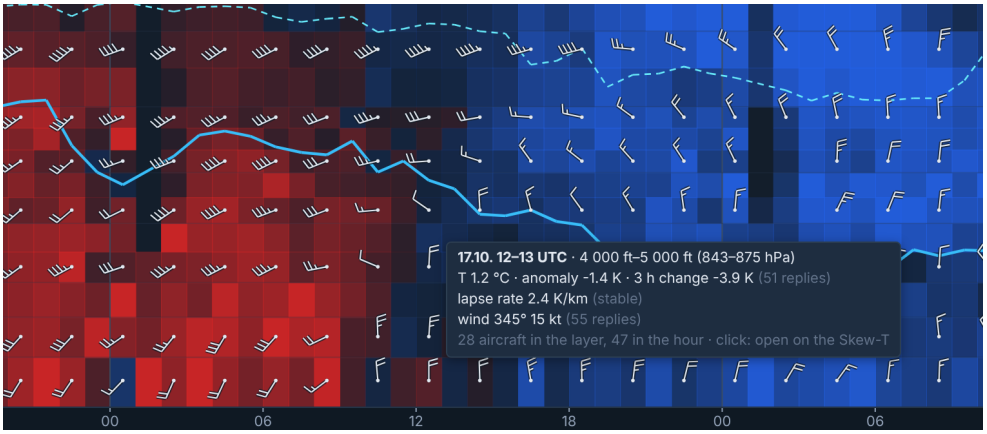


Figure 5. Tooltip on 17 October 12–13 UTC at 4 000–5 000 ft, just behind the cold front: 1.2 °C, 1.4 K colder than the period mean and 3.9 K colder than three hours earlier, stable (2.4 K/km), wind 345°/15 kt, 28 aircraft in the layer.

5.2 Temperature anomaly: air masses and fronts

The **temperature anomaly** is each layer's temperature minus the mean of the same layer over the shown period (blue colder, red warmer, full colour at ±6 K). Subtracting the layer mean removes the normal fall of temperature with height, so what remains are the air masses. Note that the reference is the *shown period*: in a 72-hour view a front is a strong contrast, in a 14-day view the same hours may be only part of a longer warm or cold spell. Change the length to change the reference.

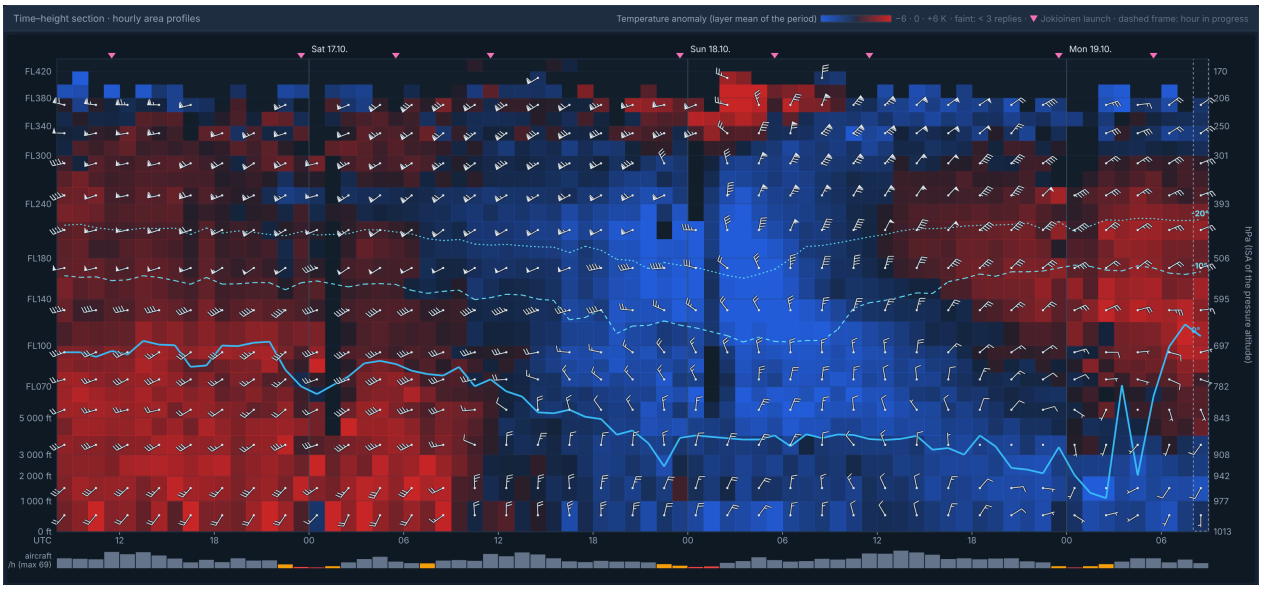


Figure 6. Temperature anomaly, 72 hours to 19 October 2026 08 UTC, with barbs and isotherms. Warm air (red) with south-westerly winds on 16–17 October; the cold front arrives near the ground at about 10 UTC on 17 October and higher up later, so the boundary rises to the right; the 0 °C line falls from about FL100 to 3 500 ft. Early on 18 October a lowered tropopause makes FL340–FL380 warm. From the afternoon of 18 October the middle troposphere warms (sinking air under the high) while the lowest 3 000 ft stay cold.

Reading the cold front of 17 October (Fig. 6). The hourly area means show the cold air arriving at different times at different heights:

- 0–4 000 ft: cooling of 4–6 K between 08 and 12 UTC, and the wind turning from south-west (244°/21 kt at 1 000 ft at 08 UTC) to north (358°/23 kt at 10 UTC);
- 6 000–10 000 ft: cooling mainly between 12 and 18 UTC, the wind turning from south-west to west and north-west between about 14 and 18 UTC;
- FL140–FL240: cooling between 14 and 22 UTC.

The boundary therefore climbs about 20 000 ft in 10–12 hours. This is the wedge of Fig. 2 seen in time: a cold front's cold air is shallow at its leading edge and deepens behind it. If the front moved at about 20 kt, 11 hours correspond to about 400 km, a slope of roughly 1:70 — within the 1:50 to 1:100 typical of cold fronts. A *warm front* would look the other way round: the warm air arrives aloft first and reaches the ground hours later, so its boundary *descends* to the right.

Troughs and the tropopause. Behind the front, early on 18 October, the layers at FL340–FL380 turn red: at FL380 the temperature rose from about −58 to −50 °C. In the cold air behind a trough the tropopause is low, and the air at these levels is now stratospheric — warmer than the tropospheric air that is normally there. A warm patch at the top of the section above cold air below is the signature of a lowered tropopause.

Sinking air under a high. From the afternoon of 18 October the QNH rose from 1031 to 1040 hPa and the middle troposphere warmed strongly: at 10 000 ft from −11 °C (08 UTC on the 18th) to −1 °C (06 UTC on the 19th), at 6 000 ft by 6–7 K, while the lowest 2 000 ft hardly changed. Air sinking in an anticyclone is compressed and warms at the dry adiabatic rate; descent of a few hundred metres to a couple of kilometres a day is typical. The warming reaches down only to the top of the cool, moist boundary layer, so a *subsidence inversion* forms between them (Section 5.5).

The daily cycle. In the lowest 1 000–3 000 ft the anomaly alternates between day and night on clear days: daytime heating mixes up to the dry thermal top, nights cool the lowest layers.

5.3 Temperature change over 3 hours: advection as it happens

The **3-hour change** is each layer's temperature minus that of the same layer three hours earlier (full colour at ±4 K). It does not depend on the period, and it shows *when* things happen. The local change of temperature has three causes:

$$\frac{\partial T}{\partial t} = \underbrace{-\vec{V}\cdot\nabla T}_{\text{advection}} + \underbrace{(\Gamma_d - \Gamma)\,w}_{\text{vertical motion}} + \underbrace{Q}_{\text{heating, cooling}}$$

Advection is warmer or colder air carried in by the wind; *vertical motion* warms sinking air and cools rising air (in proportion to how much less steep the actual lapse rate Γ is than the dry adiabatic Γ_d); *diabatic* heating and cooling are the sun, night-time radiation, condensation and evaporation. Near the ground the daily cycle dominates; above about 5 000 ft advection and vertical motion do.

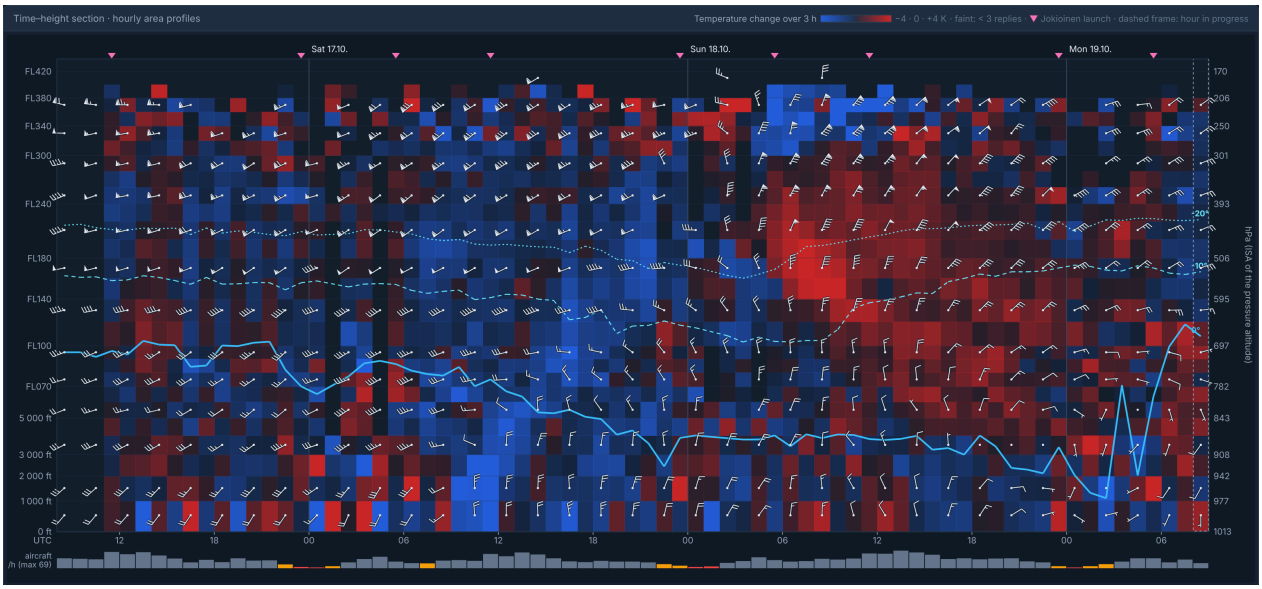


Figure 7. Temperature change over 3 hours, same period. The cold front is a blue band rising from the ground at about 10 UTC on 17 October to FL240 around midnight; the sinking air of 18 October is a broad red area at FL100–FL240 from about 05 to 15 UTC. Near the ground, alternating red and blue cells follow the day and night.

The change field is noisier than the anomaly, because a single hour's layer mean scatters by about 1 K and the change subtracts two of them. Look for coherent areas several hours and layers wide.

5.4 Wind speed: jets and the thermal wind

The **wind speed** field colours each cell in 10 kt steps, from dark blue (calm) through green and yellow to red and purple (130 kt and more). It shows the jet streams — the strongest winds, usually just below the tropopause — their height and their timing, low-level jets near the ground and calm layers.

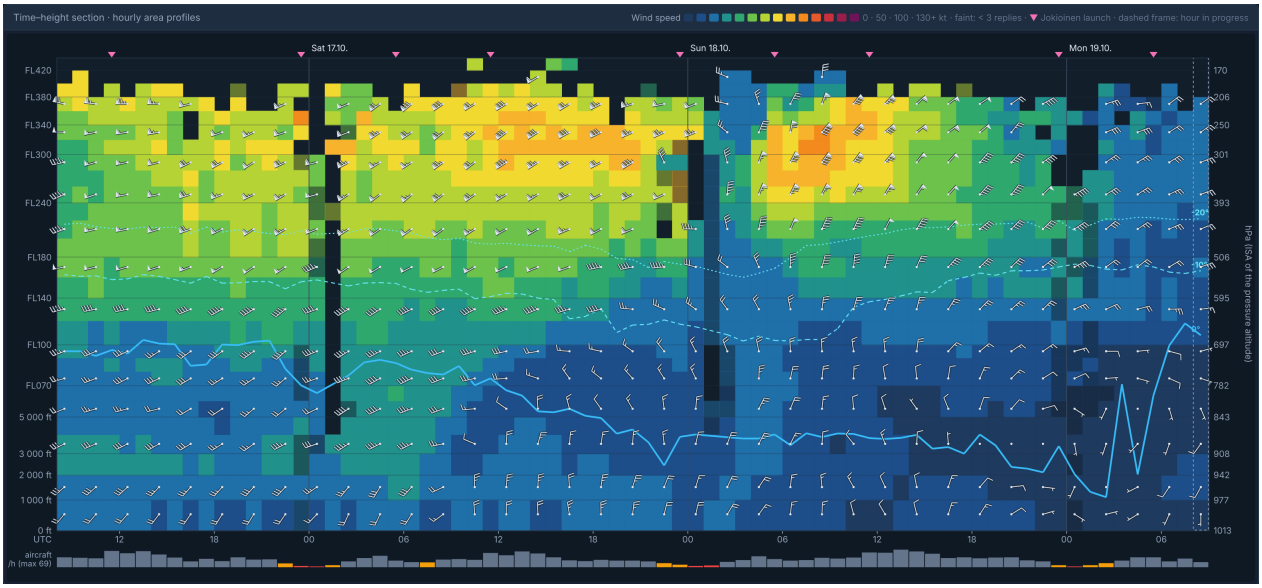


Figure 8. Wind speed, same period. A south-westerly jet of 70–95 kt at FL300–FL340 lies above the warm air and the front on 16–17 October, and a 30–35 kt south-westerly flow at 1 000–4 000 ft ahead of the front. On 18 October a north-easterly jet of up to 95 kt passes at FL300 (07–08 UTC). Under the high the winds become light at all levels.

The thermal wind. The barbs carry more than speed. Because the wind in the free atmosphere is close to geostrophic — it blows along the height contours of the pressure surfaces — its change with height is tied to the horizontal temperature gradient (the *thermal wind*): the vertical wind shear blows along the isotherms of the layer, with cold air to the left. Two practical rules follow:

- the strongest winds lie above the strongest temperature contrasts — the jet of 17 October sits above the front;

- a wind that **veers** (turns clockwise) with height means warm advection; one that **backs** (turns counter-clockwise) means cold advection. At 12 UTC on 17 October the wind backed from 004° at 1 000 ft through 345° at 4 000 ft to 257° at 8 000 ft: cold advection, exactly what the change field shows.

The Skew-T Analysis (level Advection) and the Advection row of the Indices panel turn this rule into numbers in K/h.

5.5 Lapse rate: stability and inversions

The **lapse rate** field shows how fast temperature falls with height at each layer, from the temperatures of the layers below and above, smoothed over ±1 hour (a single hour's layer means scatter by about 1 K, which over 1 000 ft would be 3 K/km of noise). The colours are classes in K/km: violet inversion (< -1), dark blue very stable (-1 to 2), light blue stable (2–5), grey normal (5–7), orange steep (7–9) and red dry-unstable (≥ 9, the dry adiabatic rate is 9.8).

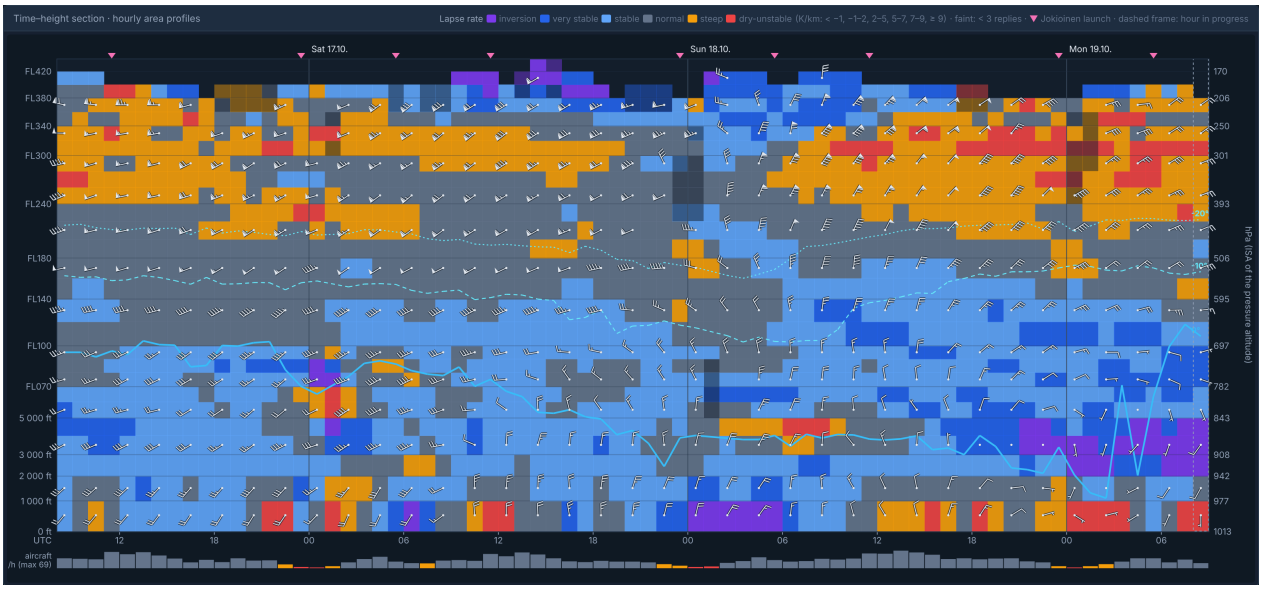


Figure 9. Lapse rate, same period. A night-time surface inversion (violet at 0–1 000 ft) in the early hours of 18 October; steep, near-adiabatic lapse rates near the ground on the sunny afternoon of 18 October; the subsidence inversion forming at 3 000–5 000 ft from the evening of 18 October; steep upper-tropospheric lapse rates below the tropopause and a very stable band at about FL380 (the tropopause). The red cells near the ground around midnight of 19 October rest on 2–8 aircraft (red bars in the strip) and are noise.

Stability decides much of what the weather does:

- **Inversions** stop vertical mixing. Below them haze, fog and low stratus are trapped; a sharp change of wind across the inversion top gives low-level wind shear. Night-time *radiation inversions* form from the ground up on clear, calm nights and are mixed away by the morning sun. *Subsidence inversions* form aloft under a high, as on 18–19 October, and can last for days. *Frontal inversions* separate the cold air under a front from the warm air above (the warm nose at 1 500 ft in Fig. 2).
- **Steep lapse rates** mean that rising air stays warmer than its surroundings: thermals near the ground, convection and showers when the air is moist, turbulence.
- The **tropopause** appears as the change from orange or grey to blue at the top of the section: the stratosphere above is very stable.

5.6 Isotherms and the freezing level

The 0, −10 and −20 °C lines are the levels most used in aviation weather: the freezing level for the type of precipitation and for icing, the 0 to −20 °C layer for supercooled water. In Fig. 6 the cold front lowers the 0 °C level from about FL100 to 3 500 ft in 12 hours and the −10 °C level by about 6 000 ft; under the high on 19 October the 0 °C line jumps between 2 000 ft and FL100 because the air is close to 0 °C over the whole lower troposphere.

5.7 Longer periods

The 7- and 14-day views show the weather regimes: a series of warm and cold air masses, the jet stream coming and going, and the gradual change of the season (Fig. 10). Barbs are best switched off at this length. The hourly archive keeps every profile, so the section can be drawn for any past period by choosing its end date; hours stored before 11 October 2026 have no aircraft count (no strip).

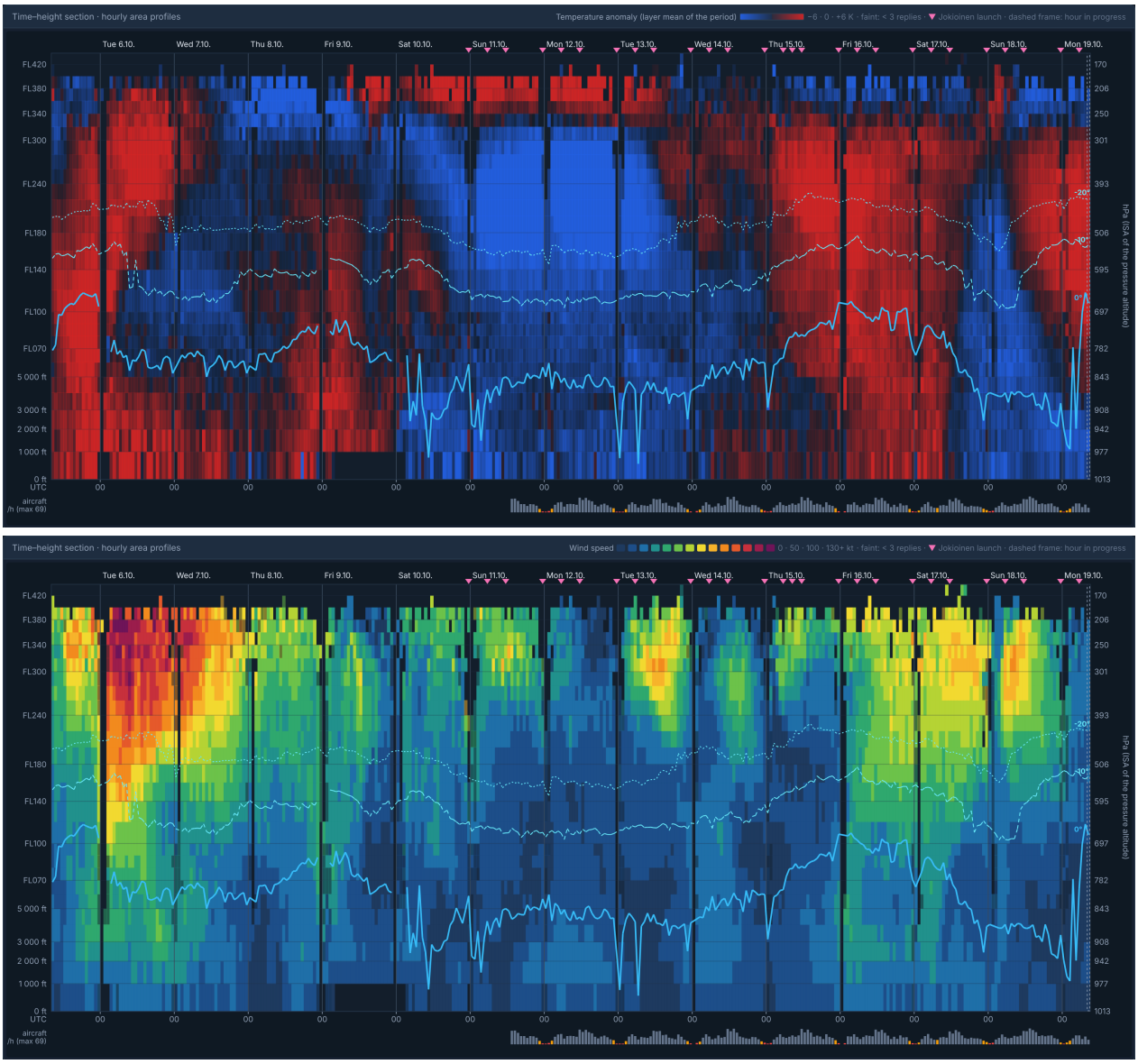


Figure 10. 14 days to 19 October 2026, temperature anomaly (top) and wind speed (bottom), barbs off. A jet of up to 145 kt (FL320, 6 October); a cold spell from 10 to 13 October with the 0 °C level near 3 000–5 000 ft and the tropopause low (warm FL340–FL380); warm air again from 14 to 17 October, the front on 17 October and the high of 18–19 October. The dark vertical lines are night hours without data.

5.8 A suggested way to use it

1. Start with 72 h, Temperature anomaly, barbs and isotherms on: find the air masses and any sloping boundaries.
2. Switch to Temperature change 3 h to time them, and check the barbs for veering or backing with height.
3. Look at Wind speed for the jet and low-level jets, and at Lapse rate for inversions and steep layers.
4. Click an interesting column to open that hour on the Skew-T, with the Indices panel and the radiosonde.
5. Before trusting a single column, check the aircraft strip and the faint cells.

6 Sonde check: how good is the aircraft profile?

The Sonde check view compares the Mode S area profile with every stored Jokioinen sounding, from a table that the hourly jobs fill by themselves. The period buttons choose 7, 30, 90 days or a year (Fig. 11).

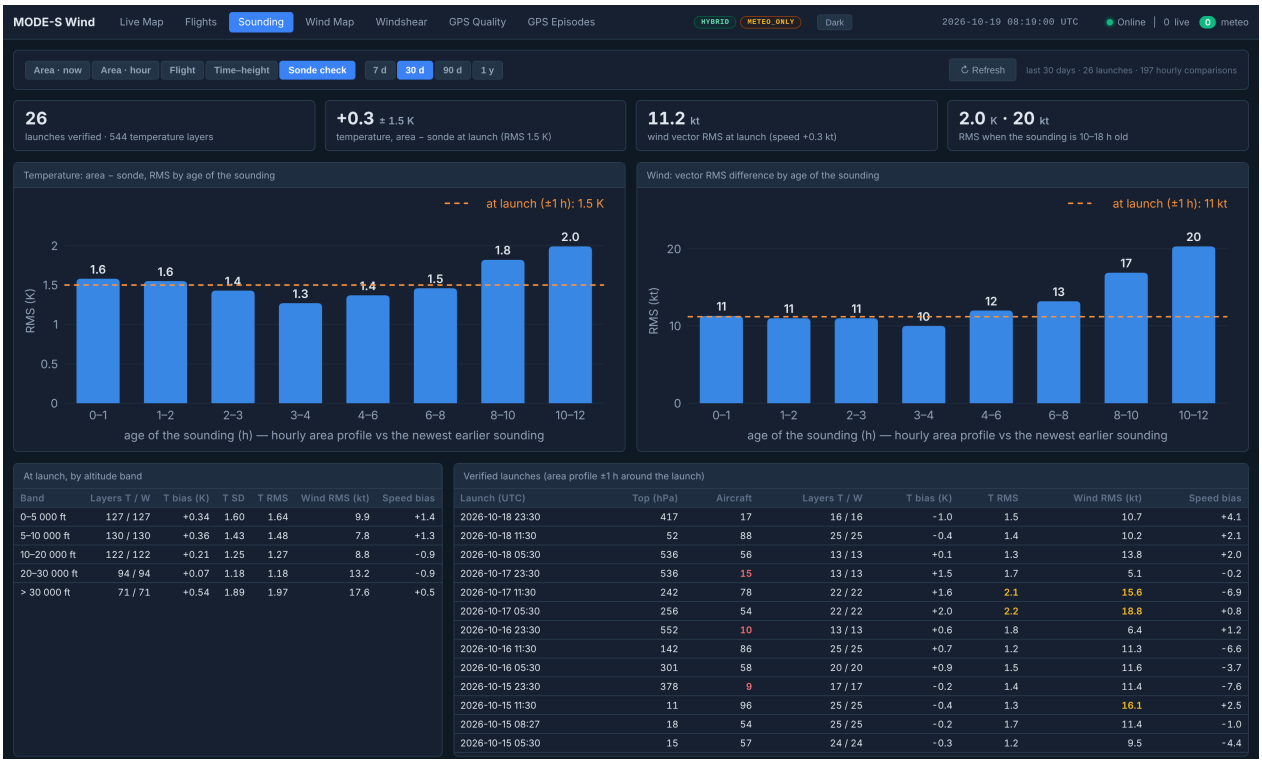


Figure 11. Sonde check, 30 days to 19 October 2026: 26 launches, temperature area − sonde $+0.3 \pm 1.5$ K, wind vector RMS difference 11 kt; the difference by the age of the sounding (bars) and by altitude band; the list of launches.

At launch. Each sounding is compared with the area profile of ± 1 hour around its launch. For temperature the page gives the *bias* (mean difference area − sonde), the *standard deviation* (scatter around the bias) and the *RMS* difference, $\sqrt{\text{bias}^2 + \text{SD}^2}$; for wind the RMS length of the vector difference and the speed bias. The table by altitude band and the list of launches show where and when the two disagree. Amber values in the list (T RMS > 2 K, wind RMS > 15 kt) and red aircraft counts (< 16, a thin night profile) point to cases worth a look. In October 2026 the two worst launches were 17 October 05:30 and 11:30 UTC, both with the cold front between Jokioinen and Helsinki (Fig. 2) — a difference in the weather, not in the measurement.

Why they can never agree perfectly. The two systems measure different things. The radiosonde is a point that drifts with the wind (tens of kilometres by the time it reaches cruise levels), measures every few metres and needs about an hour to reach 10 km. The area profile is a 300 km wide, 60-minute average, 100 km away. Part of the difference is therefore real *representativeness*: inland Jokioinen has colder nights near the ground, fronts and showers pass one site before the other, and jet cores are narrower than 300 km — which is why the wind differences are largest above 30 000 ft. The 30-day temperature bias of about $+0.3$ K is common to the whole fleet; it cannot be found by comparing aircraft with each other, only against an independent reference like this one.

By age of the sounding. Every archived hour is also compared with the newest sounding launched by the middle of that hour. The bar charts give the RMS difference by the age of that sounding: how fast the atmosphere moves away from a sounding. In October 2026 the temperature difference stayed at 1.3–1.6 K for the first 6 hours and grew to 2.0 K at 10–12 hours; the wind difference stayed at 10–12 kt for 6 hours and reached 20 kt at 10–12 hours. The first hours show the floor set by measurement and representativeness; the growth after that is the weather changing. Since soundings are made only every 6 or 12 hours (there is no 18 UTC launch), this is what a live aircraft profile adds: between the launches it follows the atmosphere, while the last sounding grows steadily older.

7 Limitations

Where aircraft fly. The profile exists only where aircraft fly: cruise levels are well sampled by day, the lowest levels mainly along the EFHK approach and departure paths, and the night hours (about 22–02 UTC) may have only a few aircraft. Watch the data badge, the red rings, the A/C column and the aircraft strip of the time–height section. Data usually end at about FL410, so a high tropopause is missed.

Averaging. The area profile is an average over 150 km and 60 minutes. A front passing through the area is smeared over several hours, thin inversions and a sharp tropopause are smoothed over a 1 000–2 000 ft layer, and jet maxima are weaker than in a point measurement.

Accuracy. Layer mean temperatures are good to about 1–1.5 K against the sonde, the lowest layers less certain because single replies at approach speeds scatter by 3–4 K; winds are good to about 10 kt RMS. Single-flight profiles carry the scatter of one airframe.

No humidity aloft. Indices that need dewpoints aloft are sonde only. The surface parcel uses the METAR dewpoint and assumes a well-mixed boundary layer; clouds above the ground, icing itself and contrail persistence cannot be derived.

Heights. The vertical scale is pressure altitude (ISA). The Indices panel computes true heights hypsometrically from the METAR surface pressure; plain temperature is used throughout (no virtual-temperature correction), which changes the 1000–500 hPa thickness by well under 1 dam in Finnish autumn air.

Not an official product. The automatic indices and analysis follow simple, documented rules on averaged data. Use them to understand the atmosphere, not for flight planning; use official forecasts and soundings for that.

8 Quick reference

Item	Meaning
Time–height section	
red / blue (anomaly)	warmer / colder than the layer's mean over the shown period (full at ±6 K)
red / blue (3-hour change)	warming / cooling over the last 3 hours (full at ±4 K)
blue → green → yellow → red → purple	wind speed in 10 kt steps (purple ≥ 130 kt)
violet / dark blue / light blue / grey / orange / red	lapse rate < −1 / −1–2 / 2–5 / 5–7 / 7–9 / ≥ 9 K/km
solid / dashed / dotted cyan line	0 / −10 / −20 °C (lowest crossing per hour)
faint cell, dark cell	fewer than 3 replies, no data
pink triangle	Jokioinen radiosonde launch
red / amber bar in the strip	fewer than 8 / 20 aircraft in the hour
dashed frame	hour in progress (computed live)
Indices panel (amber / red)	
CAPE	≥ 100 / ≥ 1 000 J/kg
Lifted Index; Showalter	≤ −1 / ≤ −5; ≤ 0 / ≤ −3
K index; Total Totals; Vertical Totals	≥ 26 / ≥ 31; ≥ 44 / ≥ 50; ≥ 30
lapse rates	≥ 7 / ≥ 9 K/km
shear 0–1 km; 0–6 km	≥ 20 kt; ≥ 40 kt (amber)
CCL · T conv amber	the surface has reached the convective temperature
Level table and Sonde check	
red A/C	layer measured by only 1–2 aircraft
amber ΔT; ΔV	more than 2 K; more than 10 kt from the sonde
amber T RMS; wind RMS; red aircraft	> 2 K; > 15 kt; < 16 aircraft behind a launch comparison

A Formulas

- Mach/TAS temperature: $T = 288.15\,(\mathrm{TAS}/(661.47\,M))^2$ K (TAS in kt).
- ISA pressure of pressure altitude h (ft): $p = 1013.25\,(1 - 6.8756\times10^{-6}\,h)^{5.2559}$ hPa below 36 089 ft, $p = 226.32\exp\big(-(h-36089)/20806\big)$ above.
- Hypsometric thickness: $\Delta z = \dfrac{R\bar{T}}{g}\ln\dfrac{p_1}{p_2}$, $R = 287.04$ J/(kg K), $g = 9.80665$ m/s^2. For 1000–500 hPa: $\bar{T} = g\,\Delta z/(R\ln 2)$, so 1 dam ≈ 0.49 K.
- Lapse rate between two levels: $\Gamma = -(T_2 - T_1)/(z_2 - z_1)$; time–height: central difference between the layers below and above, with temperatures smoothed over ±1 h (weights 0.5 / 1 / 0.5 times the number of replies).
- Anomaly: $T'(t,z) = T(t,z) - \overline{T}(z)$, the mean over the shown period weighted by replies; 3-hour change: $T(t,z) - T(t-3\,\mathrm{h},z)$.
- Local temperature change: $\partial T/\partial t = -\vec{V}\cdot\nabla T + (\Gamma_d - \Gamma)\,w + Q$, $\Gamma_d = g/c_p = 9.8$ K/km.
- Geostrophic wind: $\vec{V}_g = \dfrac{g}{f}\,\hat{k}\times\nabla_p z$, $f = 2\Omega\sin\varphi \approx 1.26\times10^{-4}$ s^{-1} at 60°N.
- Thermal wind: $\vec{V}_T = \vec{V}_g(p_2) - \vec{V}_g(p_1) = \dfrac{R}{f}\ln\dfrac{p_1}{p_2}\,\hat{k}\times\nabla\bar{T}$; thermal advection $-\vec{V}\cdot\nabla\bar{T} = -\dfrac{f}{R\ln(p_1/p_2)}\,(u_1 v_2 - v_1 u_2)$ (positive = warm advection, wind veering with height).
- LCL (Bolton 1980): $T_L = 1\big/\big(1/(T_d - 56) + \ln(T/T_d)/800\big) + 56$ (K), $p_L = p_0\,(T_L/T)^{c_p/R}$.
- CAPE $= R\int_{p_{EL}}^{p_{LFC}} (T_p - T_e)\,d\ln p$; CIN the negative part below the LFC; Lifted Index $= T_e(500) - T_p(500)$; Showalter the same for a parcel from 850 hPa.
- K index $= (T_{850} - T_{500}) + T_{d,850} - (T_{700} - T_{d,700})$; Vertical Totals $= T_{850} - T_{500}$; Total Totals $=$ VT $+ (T_{d,850} - T_{500})$.
- Precipitable water: $W = \dfrac{1}{g}\int_{p_{300}}^{p_s} r\,dp$, r the mixing ratio.
- WMO tropopause: the lowest level above 5 km at which the lapse rate falls to ≤ 2 K/km, provided the average lapse rate to every level within the next 2 km does not exceed 2 K/km.
- Verification: bias $= \overline{\Delta T}$, RMS $= \sqrt{\overline{\Delta T^2}} = \sqrt{\mathrm{bias}^2 + \mathrm{SD}^2}$; wind vector RMS $= \sqrt{\overline{|\vec{V}_a - \vec{V}_s|^2}}$.

Written with the assistance of Claude (Anthropic). Screenshots: MODE-S Wind, dark theme, with the database and the Jokioinen soundings (FMI open data, CC BY 4.0) of 5–19 October 2026. The surface values in Fig. 1 are those of the METAR stored with the hourly profile of 19 October 07 UTC; the page was shown as at 19 October 2026 08:19 UTC.